

Dhan Michael Pagulayan

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EDUCATION

University of the Philippines-Diliman

Bachelor of Science in Mechanical Engineering — GPA: 1.3631 (University Scholar)

Quezon City, Metro Manila

Expected May 2029

EXPERIENCE

Obeikan Liquid Packaging

April 2025

Work Immersion Student

Riyadh, Saudi Arabia

- Completed an intensive **80-hour immersion** across Production, Quality Control, Tooling, and Maintenance departments at a leading Middle East liquid packaging manufacturer, gaining end-to-end exposure to the full production pipeline — from blank coating and printing to die-cutting, flame sealing, and final packaging
- Performed hands-on quality inspections and identified defective production stacks through visual analysis, while assisting technicians in reconfiguring cutting tools for different blank specifications

PROJECTS

Hydraulic Operated Arm Design and Assembly | *Fusion 360*

July 2026

- Designed a 3D-printable hydraulic arm assembly in ABS plastic around a given syringe component, achieving **5 degrees of freedom** while meeting functional and printability constraints
- Engineered and assembled all supporting components into a fully functional mechanical system, balancing design decisions against 3D-printing limitations such as wall thickness and tolerances

Steel Grill Fabrication | *Wood, Machine, and Sheet Metal Shop*

July 2026

- Co-designed and fabricated a steel grill as part of a **5-person team**, from an original engineering drawing approved by faculty, combining an angle bar base, sheet-metal body, and spot-welded stainless steel rod mesh
- Designed a hand-free mesh removal mechanism and crafted hardwood handles as part of a team, translating the drawing into a functional physical product

Mixer Housing and Handle | *Fusion 360*

July 2026

- Reverse-engineered a mixer housing, handle, beater, and front/back grills from general measurements and a provided handle file, achieving fit within **1 percent tolerance of original dimensions**
- Resolved interference across **8 screw and fastener points** while integrating pre-supplied internal components into the housing design

Adjustable Tap Wrench Assembly | *Fusion 360*

July 2026

- Assembled a multi-component adjustable tap wrench from provided engineering drawings, applying precise mechanical constraints in Fusion 360 to ensure correct part relationships across all components
- Engineered the sliding jaw mechanism to accommodate multiple tap sizes, ensuring smooth adjustment and secure clamping under torque

Automotive Wheel Design | *Fusion 360*

December 2025

- Independently designed a life-size **18-inch, 9J** automotive wheel and brake assembly in Fusion 360 as a personal design challenge, modeling curved rim geometry inspired by Porsche and BMW styling
- Detailed **5 individual components** — rim, tire, caliper, disc, and brake pad — with realistic tread, surface texturing, and material assignments

TECHNICAL SKILLS

CAD and Design: Fusion 360, 3D Parametric Modeling, Mechanical Design, Mechanical Assembly Design, CAD Rendering

Engineering: Engineering Drawing, Dimensioning, Mechanism Design

Languages: Basic Python, Basic Matlab